Classical Mechanics by Taylor

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1 Newton's Laws of Motion

1.1 Space and Time

Problem 8

1. Use the definition (1.7) to prove that the scalar product is distributive, that is $\vec{r} \cdot (\vec{u} + \vec{v}) = \vec{r} \cdot \vec{u} + \vec{r} \cdot \vec{v}$.
2. If $\vec{r}$ and $\vec{s}$ are vectors that depend on time, prove that the product rule for differentiating products applies to $\vec{r} \cdot \vec{s}$, that is, that
$$\frac{d}{ds}(\vec{r} \cdot \vec{s}) = \vec{r} \cdot \frac{d\vec{s}}{dt} + \frac{d\vec{r}}{dt} \cdot \vec{s}.$$

Proof. We have

$$\vec{r} \cdot (\vec{u} + \vec{v}) = \sum_{i=1}^{3} r_i(u_i + v_i) = \sum_{i=1}^{3}(r_i u_i + r_i v_i) = \sum_{i=1}^{3} r_i u_i + \sum_{i=1}^{3} r_i v_i = \vec{r} \cdot \vec{u} + \vec{r} \cdot \vec{v}.$$

∎

Proof. Suppose $\vec{r}$ and $\vec{s}$ depend on time t. Then

$$\frac{d}{dt}(\vec{r} \cdot \vec{s}) = \frac{d}{dt}\left(\sum_{i=1}^{3} r_i s_i\right) = \sum_{i=1}^{3} \frac{d}{dt}(r_i s_i) = \sum_{i=1}^{3}\left(\frac{dr_i}{dt} s_i + r_i \frac{ds_i}{dt}\right) = \frac{d\vec{r}}{dt} \cdot \vec{s} + \vec{r} \cdot \frac{d\vec{s}}{dt}.$$

∎

Problem 10

A particle moves in a circle (center O and radius R) with constant angular velocity ω counterclockwise. The circle lies in the xy plane and the particle is on the x axis at time $t = 0$. Show that the particle's position is given by
$$\vec{r}(t) = \hat{x} R \cos(\omega t) + \hat{y} R \sin(\omega t).$$

Find the particle's velocity and acceleration. What are the magnitude and direction of the acceleration? Relate your results to well-known properties of uniform circular motion.

Proof. We begin with the position of a point P lying on the unit circle

$$P = (\cos x, \sin x).$$

Now we scale the radius of the circle by R such that the point P lies R away from O

$$P = (R\cos x, R\sin x).$$

We let $x = \omega t$ be the position at time t for angular velocity ω, giving

$$\vec{r}(t) = (R\cos(\omega t), R\sin(\omega t)) = \hat{x}R\cos(\omega t) + \hat{y}R\sin(\omega t).$$

The velocity vector is obtained by differentiating

$$\vec{v}(t) = \frac{d\vec{r}}{dt} = (-R\omega\sin(\omega t), R\omega\cos(\omega t)).$$

The acceleration vector is obtained by differentiating again

$$\vec{a}(t) = \frac{d\vec{v}}{dt} = (-R\omega^2\cos(\omega t), -R\omega^2\sin(\omega t)) = -\omega^2\vec{r}(t).$$

The force vector is pulling the point towards the center of the circle. The acceleration vector is in the same direction as the force vector, so it points toward the origin. Thus the magnitude is

$$\|\vec{a}(t)\| = \omega^2 R,$$

and the direction is toward the center (opposite to the position vector $\vec{r}(t)$). ∎

Solution: Imagine you have a string with a ball tied to the end. As you increase the velocity of the ball you feel a greater force as the ball tries to escape its path. This corresponds to the ω^2 term in the force magnitude formula. Now imagine you half the length of the string and spin with equivalent velocity. You will feel less force corresponding to $\frac{1}{R}$ in the magnitude $\omega^2 R$.

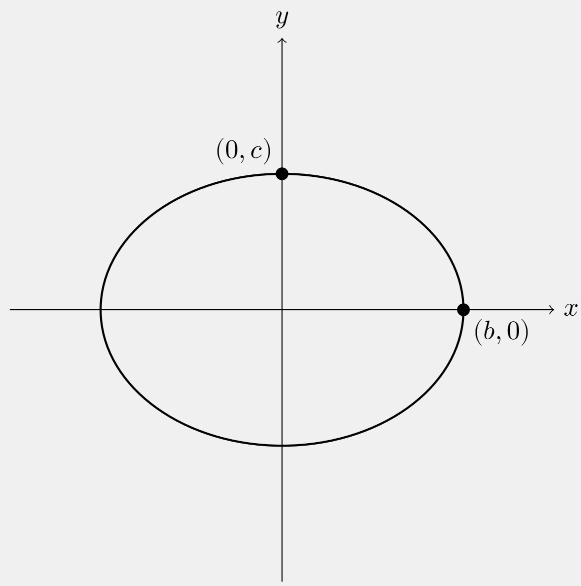
Problem 11

The position of a moving particle is given as a function of time t to be

$$\vec{r}(t) = \hat{x}b\cos(\omega t) + \hat{y}c\sin(\omega t),$$

where b, c and ω are constants. Describe the particles orbit.

Proof. This is the standard form for an ellipse. At $\omega t = 0$ we have $r(t) = (b, 0)$ and at $\omega t = \pi/2$ we have $r(t) = (0, c)$. The following is the shape.



The particles position changes with angular velocity ω. ∎

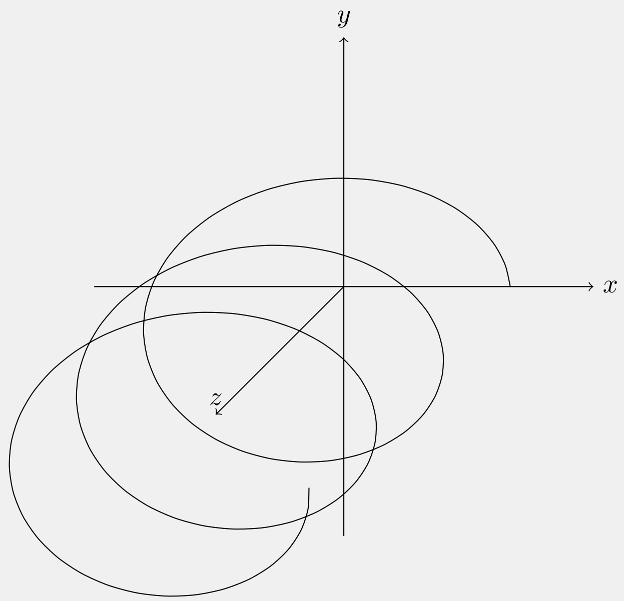
Problem 12

The position of a moving particle given as a function of time t to be

$$\vec{r}(t) = \hat{x}b\cos(\omega t) + \hat{y}c\sin(\omega t) + \hat{z}v_o t,$$

where b, c, v_o and w are constants. Describe the particle's orbit.

Proof. When $v_0 = 0$ we have Problem 11. Therefore suppose $v_0 \neq 0$. Take z to be the 'height' axis. If $v_0 > 0$ ($v_0 < 0$) the particle continuously increases (decreases) in height over time. If we take two velocities equal but opposite in sign then the path traced is equivalent but the direction they go in terms of height changes. The following is the shape.



∎

Problem 17

1. Prove that the vector product $\vec{r} \times \vec{s}$ as defined by (1.9) is distributive, that is, that $\vec{r} \times (\vec{u} + \vec{v}) = (\vec{r} \times \vec{u}) + (\vec{r} \times \vec{v})$.
2. Prove the product rule
$$\frac{d}{dt}(\vec{r} \times \vec{s}) = \vec{r} \times \frac{d\vec{s}}{dt} + \frac{d\vec{r}}{dt} \times \vec{s}.$$

Be careful with the order of the factors.

Proof. We have
$$\vec{r} \times (\vec{u} + \vec{v}) = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ r_x & r_y & r_z \\ u_x + v_x & u_y + v_y & u_z + v_z \end{vmatrix} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ r_x & r_y & r_z \\ u_x & u_y & u_z \end{vmatrix} + \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ r_x & r_y & r_z \\ v_x & v_y & v_z \end{vmatrix}.$$

Thus
$$\vec{r} \times (\vec{u} + \vec{v}) = (\vec{r} \times \vec{u}) + (\vec{r} \times \vec{v}).$$

∎

Proof. We have
$$\vec{r} \times \vec{s} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ r_x & r_y & r_z \\ s_x & s_y & s_z \end{vmatrix}.$$

Differentiating componentwise
$$\frac{d}{dt}(\vec{r} \times \vec{s}) = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \dot{r}_x & \dot{r}_y & \dot{r}_z \\ s_x & s_y & s_z \end{vmatrix} + \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ r_x & r_y & r_z \\ \dot{s}_x & \dot{s}_y & \dot{s}_z \end{vmatrix}.$$

Therefore
$$\frac{d}{dt}(\vec{r} \times \vec{s}) = \frac{d\vec{r}}{dt} \times \vec{s} + \vec{r} \times \frac{d\vec{s}}{dt}.$$

∎

1.2 Newton's First and Seconds Laws; Intertial Frames

Problem 24

Find the general solution of the first-order equation $df/dt = f$ for an unknown function $f(t)$.

Proof. Let $f(t) = Ce^t$ where C is an arbitrary constant. Then
$$\frac{df}{dt} = \frac{d}{dt}(Ce^t) = C\frac{d}{dt}e^t = Ce^t = f(t).$$

∎

Problem 25

Answer the same question as before, but for the differential equation $df/dt = -3f$.

Proof. Let $f(t) = Ce^{-3t}$ where C is an arbitrary constant. Then

$$\frac{df}{dt} = \frac{d}{dt}(Ce^{-3t}) = C(-3)e^{-3t} = -3Ce^{-3t} = -3f(t).$$

∎

Problem 26

Consider the following: I am standing on a level floor at the origin of an inertial reference frame S and kick a frictionless puck due north across the floor.

1. Write down the x and y coordinates of the puck as functions of time as seen from my inertial frame.
2. Now consider two more observers, the first at rest in a frame S' that travels with constant velocity v due east relative to S, the second at rest in a frame S'' that travels with constant acceleration due east relative to S. Find the coordinates x', y' of the puck and describe the puck's path as seen from S'.
3. Do the same for S''. Which of the frames is inertial.

Solution (a): The position of the puck is $r(t) = (0, v't)$ where v' is the velocity of the puck. This is certainly a straight line, and since there are no forces acting on the puck, Newton's first law holds in S.

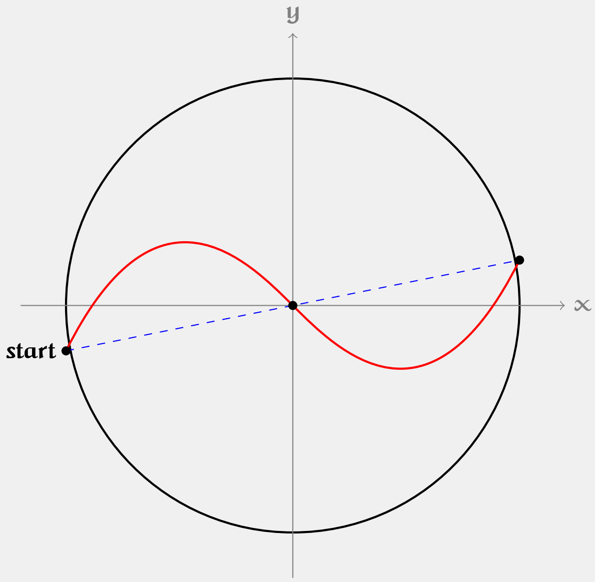
Solution (b): S' will observe the puck getting farther away at a constant velocity v. Thus the position of the puck relative to S' is $r'(t) = (-vt, v't)$. This is certainly a straight line, and since there are no forces acting on the puck, Newton's first law holds in S'.

Solution (c): S'' will observe the puck moving with an accelerating motion at rate a. Thus the position of the puck relative to S'' is $r''(t) = \left(-\frac{1}{2}at^2 - vt,\, v't\right)$. This is indeed not a straight line, and since there are no forces acting on the puck, Newton's first law does not hold in S''.

Problem 27

I am standing on the ground (which we shall take to be an inertial frame) beside a perfectly flat horizontal turntable, rotating with constant angular velocity ω. I lean over and shove a frictionless puck so that it slides across the turntable, straight through the center. The puck is subject to zero net force and, as seen from my inertial frame, travels in a straight line. Describe the puck's path as observed by someone sitting at rest on the turntable. This requires careful thought, but you should be able to get a qualitative picture.

Solution: Place the noninertial reference frame at the center of the turntable with angular velocity ω. Place the inertial reference frame at the center of the turntable with the same direction as the noninertial frame but without having an angular velocity. Have the table spin counterclockwise and suppose the table has turned π radians when the puck reaches the center. We find that the two reference frames agree on the position of the puck at the start, middle, and end.



1.3 The Third Law and Conservation of Momentum

Problem 30

Two objects of masses m_1 and m_2 are subject to no external forces. Object 1 is traveling with velocity $\vec{v}$ when it collides with a stationary object 2. The two objects stick together and move off with common velocity $\vec{v}'$. Use conservation of momentum to find v' in terms of $\vec{v}, m_1$, and m_2.

Proof. The final momentum is

$$\vec{p}' = (m_1 + m_2)\vec{v}' = m_1\vec{v} + m_2 0 = m_1\vec{v}.$$

So

$$\vec{v}' = \frac{m_1\vec{v}}{m_1 + m_2}.$$

∎

Problem 31

Prove that if the law of conservation of momentum applies to every possible group of particles, then the interparticle forces must obey the third law.

Proof. Consider two particles $1, 2$ and write

$$p_1 = m_1\vec{v}_1 \text{ and } p_2 = m_2\vec{v}_2.$$

Let $p = p_1 + p_2 = m_1\vec{v}_1 + m_2\vec{v}_2$. Taking derivatives we find

$$0 = m_1\vec{a}_1 + m_2\vec{a}_2 = \vec{F}_{12} + \vec{F}_{21}.$$

Thus $\vec{F}_{12} = -\vec{F}_{21}$. ∎

1.4 Newton's Second Law in Cartesian Coordinates

Problem 35

A golf ball is hit from ground level with speed v_0 in a direction that is due east and at an angle θ above the horizontal. Neglecting air resistance, use Newton'S second law (1.35) to find the position as a function of time, using coordinates with x measured east, y north, and z vertically up. Find the time for the golf ball to return to the ground and how far it travels in that time.

Proof. We first analyze the motion in the z-direction. Naturally, we place our coordinate system at the golf ball when it is on the ground. Now, we know

$$F_z = ma_z.$$

Since the only force is gravity, $a_z = -g$. We integrate twice to find

$$\frac{1}{2}(-g)t^2 + v_0 \sin(\theta)\, t.$$

For the x direction there is no acceleration and the solution is simply

$$x(t) = v_0 \cos(\theta)\, t.$$

For the y-direction, there is no motion, so

$$y(t) = 0.$$

Combining these we find the position function

$$r(t) = (v_0 \cos(\theta)\, t,\ 0,\ \tfrac{1}{2}(-g)t^2 + v_0 \sin(\theta)\, t).$$

To find the time until it hits the ground we set the z position to 0 and solve

$$\frac{1}{2}(-g)t^2 + v_0 \sin(\theta)\, t = 0.$$

This is a quadratic with two solutions: $t = 0$ and $t = \frac{2v_0 \sin(\theta)}{g}$. Discarding $t = 0$, we take

$$t' = \frac{2v_0 \sin(\theta)}{g}.$$

Then the distance the ball travels is simply

$$x(t') = v_0 \cos(\theta)\, t' = \frac{2v_0^2 \sin(\theta) \cos(\theta)}{g}.$$

∎

Problem 36

A plane, which is flying horizontally at a constant speed v_o and a height h above the sea, must drop a bundle of supplies to a castaway on a small raft.

1. Write down Newton's second law for the bundle as it falls from the plane, assuming you can neglect air resistance. Solve your equations to give the bundle's position in flight as a function of time t.
2. How far before the raft (measured horizontally) must the pilot drop the bundle if it is to hit the raft? What is this distance if $v_o = 50m/s$, $h = 100m$, and $g \approx 10m/s^2$?
3. Within what interval of time ($\pm\nabla t$) must the pilot drop the bundle if it is to land withing $\pm 10m$ of the

raft?

Proof. In the y-direction we have

$$F_y = ma_y = -mg.$$

so $a_y = -g$. Integrating once gives velocity

$$v_y(t) = v_{y0} - gt.$$

Since the bundle is dropped, $v_{y0} = 0$, so

$$v_y(t) = -gt.$$

Integrating again we find position

$$y(t) = -\frac{1}{2}gt^2 + v_{y0}t + y_0.$$

We have $y_0 = h$ and $v_{y0} = 0$, thus

$$y(t) = h - \frac{1}{2}gt^2.$$

In the x-direction there is no force, so

$$x(t) = v_0 t.$$

∎

Proof. Let r_r be the position of the raft. We require

$$r_r = x(t) = v_0 t.$$

Thus we need

$$t = \frac{r_r}{v_0}.$$

Since the bundle is dropped, $v_{y0} = 0$ and $y_0 = h$. The bundle hits the sea when $y(t) = 0$, so

$$0 = h - \frac{1}{2}gt^2 \quad \Rightarrow \quad t = \sqrt{\frac{2h}{g}}.$$

Therefore

$$\frac{r_r}{v_0} = \sqrt{\frac{2h}{g}}.$$

Thus the horizontal distance is

$$r_r = v_0\sqrt{\frac{2h}{g}}.$$

∎

Solution (b): Suppose $v_0 = 50\,\mathrm{m/s}$, $h = 100\,\mathrm{m}$, and $g \approx 10\,\mathrm{m/s^2}$. Then

$$r_r = v_0\sqrt{\frac{2h}{g}} = 50\sqrt{\frac{2(100)}{10}} = 50\sqrt{20} \approx 224\,\mathrm{m}.$$

Proof. From the horizontal motion we have

$$x(t) = v_0 t.$$

Taking differentials,

$$\Delta x = v_0 \Delta t.$$

We require the landing accuracy

$$|\Delta x| \leq 10\,\mathbf{m},$$

so

$$|\Delta t| \leq \frac{10}{v_0}.$$

Thus the allowed release-time interval is

$$t = \sqrt{\frac{2h}{g}} \pm \frac{10}{v_0}.$$

∎

Problem 37

A student kicks a frictionless puck with initial speed v_o, so that it slides straight up a plane that is inclined at an angle θ above the horizontal.

1. Write down Newton's second law for the puck and solve to give its position as a function of time.
2. How long will the puck take to return to its starting point.

Proof. We ignore friction here. Take the y-axis along the incline. In the y direction we have

$$F_y = ma_y = -mg\sin(\theta).$$

Mass cancels, so

$$a_y = -g\sin(\theta).$$

Integrating twice we find

$$y(t) = v_0 t - \frac{1}{2}g\sin(\theta)t^2.$$

The puck will intersect the origin twice, once at the start and once on its way back down. Thus we set $y(t) = 0$ and ignore the $t = 0$ root to find

$$t = \frac{2v_0}{g\sin(\theta)}.$$

∎

Problem 38

You lay a rectangular board on the horizontal floor and then tilt the board about one edge until it slopes at angle θ with the horizontal. Choose your origin at one of the two corners that touch the floor, the x axis pointing along the bottom edge of the board, the y axis pointing up the slope, and the z axis normal to the board. You now kick a frictionless puck that is resting at O so that it slides across the board with initial velocity $(v_{ox}, v_{oy}, 0)$. Write down Newton's second law using the given coordinates and then find how long the puck takes to return to the floor level and how far it is from O when it does so.

Proof. We ignore friction here. We have

$$F_y = ma_y = -mg\sin(\theta), \qquad F_z = -mg\cos\theta, \qquad F_x = 0.$$

Mass cancels and integrating twice we find

$$r_y(t) = v_{oy}t - \frac{1}{2}g\sin(\theta)t^2.$$

Then we set $r_y(t) = 0$ and find

$$t = \frac{2v_{oy}}{g\sin\theta}.$$

The distance from the origin when it hits the floor is

$$r_x(t) = v_{ox}t = v_{ox}\frac{2v_{oy}}{g\sin\theta}.$$

$\blacksquare$

Problem 39

A ball is thrown with initial speed v_o up an inclined plane. The plane is inclined at an angle ϕ above the horizontal, and the ball's initial velocity is at an angle θ above the plane. Choose axes with x measured up the slop, y normal to the slope, and z across it. Write down newton's second law using these axes and find the ball's position as a function of time. Show that the ball lands a distance $R = 2v_o^2 \sin\theta\cos(\theta+\phi)/(g\sin^2\phi)$ from its launch point. Show that for given v_o and ϕ, the maximum possible range up the inclined plane is $R_{max} = v_o^2/[g(1+\sin\phi)]$.

Proof. We first identify the x, y components of the gravity force vector in our chosen coordinate system. Draw the gravity force vector such that it makes a right triangle with the x, y axes. Now the plane is at angle ψ with the horizontal. Label the triangle PQR where $\angle PQR = \pi/2$. Then $\angle QRP = \pi/2 - \psi$ (we assume here $\psi \leq 90$). Thus $\angle RPQ = \psi$. Thus, in our chosen coordinate system, the components of gravity are

$$\mathbf{g} = (-g\sin\psi,\ -g\cos\psi).$$

We now identify the x, y components of our velocity vector in our chosen coordinate system. Draw the initial velocity vector such that it makes an angle θ with the x-axis (along the incline). Since the axes are chosen with x along the slope and y normal to the slope, we resolve the velocity into components of a right triangle with the x, y axes. Label the triangle PQR where $\angle PQR = \pi/2$. Then the velocity vector makes angle θ with the x-axis, so $\angle RPQ = \theta$. Thus the components in our chosen coordinate system are

$$\boldsymbol{v}(0) = (v_0\cos\theta,\ v_0\sin\theta).$$

Integrating twice in the x, y axes gives the position function

$$r(t) = \left(v_0\cos\theta\, t - \frac{1}{2}g\sin\psi\, t^2,\ v_0\sin\theta\, t - \frac{1}{2}g\cos\psi\, t^2\right).$$

To find the intersection of the point with the plane we set the y-component equal to zero, and noting that the plane corresponds to $y = 0$, we ignore the root at $t = 0$ and find

$$t = \frac{2v_0\sin\theta}{g\cos\psi}.$$

We can plug t into $r(t)$ to get the position of the ball at its intersection with the plane

To get the maximum distance up the plane at a fixed v_0, ψ, we analyze the x component of the position function.

$$x(t) = v_0\cos\theta\, t - \frac{1}{2}g\sin\psi\, t^2.$$

The range along the plane is $R = x(t)$, so substituting the time of flight

$$t = \frac{2v_0\sin\theta}{g\cos\psi},$$

we obtain

$$R = v_0\cos\theta\left(\frac{2v_0\sin\theta}{g\cos\psi}\right) - \frac{1}{2}g\sin\psi\left(\frac{2v_0\sin\theta}{g\cos\psi}\right)^2.$$

Maximizing with respect to θ, we find the maximum range to be

$$R_{\max} = \frac{v_0^2}{g(1+\sin\phi)}.$$

∎

1.5 Two-Dimensional Polar Coordinates

Problem 41

An astronaut in gravity-free space is twirling a mass m on the end of a string of length R in a circle, with constant velocity ω. Write down Newton's second law (1.48) in polar coordinates and find the tension in the string.

Proof. In the the radial direction

$$F_r = ma_r = m\left(\frac{d^2r}{dt^2} - r\left(\frac{d\theta}{dt}\right)^2\right).$$

Since $\frac{d^2r}{dt^2} = 0$ and $r = R$, this gives

$$F_r = -mR\omega^2.$$

But the only radial force is the tension T, so

$$F_r = -T.$$

Therefore,

$$T = mR\omega^2.$$

∎

Problem 42

Prove that the transformations from rectangular to polar coordinates and vice versa are given by the four equations (1.37). Explain why the equation for ψ is not quite complete and give a complete version.

Problem 43

1. Prove that the unit vector $\hat{r}$ of two-dimensional polar coordinates is equal to

$$\hat{r} = \hat{x}\cos\psi + \hat{y}\sin\psi,$$

   and find a corresponding expression for $\hat{psi}$.
2. Assuming that ψ depends on time t, differentiate your answers in part (a) to give an alternative proof of the results (1.42) and (1.46) for the time derivatives $\hat{r}$ and $\hat{\psi}$.

Problem 44

Verify by direct substitution that the function $\psi(t) = A\sin(\omega t) + B\cos(\omega t)$ of (1.56) is a solution of the second-order differential equation (1.55), $\frac{d}{dt^2}\psi = -\omega^2\psi$.

Problem 45

Prove that if $\vec{v}(t)$ is any vector that depends on time (for example the velocity of a moving particle) but which has a constant magnitutde, then $\frac{d}{dt}\vec{t}$ is orthogonal to $\vec{v}(t)$. Prove the converse that if $\frac{d}{dt}\vec{v}(t)$ is orthogonal to $\vec{v}(t)$, then $|\vec{v}(t)|$ is constant.

Problem 46

Consider the experiment of Problem 1.27, in which a frictionless puck is slid straight across a rotating turntable through the center O.

1. Write down the polar coordinates r, ψ of hte puck as functions of time, as measured in the inertial frame S of an observer on the ground. (Assume that the puck was launched along the axis $\psi = 0$ at $t = 0$)
2. Now write down the polar coordinates r', ψ' of the puck as measured by an observer (frame S') at rest on the turntable. (Choose these coordinates so that ψ and ψ' coincide at $t = 0$.) Describe and sketch the

path seen by this second observer. Is the frame S' inertial?

Problem 47

Let the position of a point P in three dimensions be given by the vector $\vec{r} = (x, y, z)$ in rectangular (or Cartesian) coordinates. The same position can be specified by cylindrical polar coordinates, ρ, ψ, z, which are defined as follows: Let P' denote the projection of P onto the xy plane; that is, P' has Cartesian coordinates $(x, y, 0)$. Then ρ and ψ are defined as the two-dimensional polar coordinates of P' in the xy plane, while z is the third Cartesian coordinate, unchanged.

1. Make a sketch to illustrate the three cylindrical coordinates. Give expressions for ρ, ψ, z in terms of the Cartesian coordinates x, y, z. Explain in words what ρ is ("ρ is the distance of P from..."). Explain why this use of r is unfortunate.
2. Describe the three unit vectors $\hat{\rho}, \hat{\psi}, \hat{z}$ and write the expansion of the position vector r in terms of these vectors.
3. Differentiate your last answer twice to find the cylindrical components of the acceleration $\vec{a} = \frac{d}{dt^2}\vec{r}$ of the particle. To do this, you will need to know the time derivative of $\hat{\rho}$ and $\hat{\psi}$. You could get these from corresponding results (1.42) and (1.46), or you could derive them directly as in Problem 1.48.

Problem 48

Find expressions for the unit vectors $\hat{\rho}, \hat{\psi}$ and $\hat{z}$ of cylindrical polar coordiantes (Problem 1.47) in terms of the Cartesian $\hat{x}, \hat{y}, \hat{z}$. Differentiate these expressions with respect to time to find $\frac{d\hat{\rho}}{dt}$, $\frac{d\hat{\psi}}{dt}$, and $\frac{d\hat{z}}{dt}$.

Problem 49

Imagine two concentric cylinders, centered on the vertical z axis, with radii $R \pm \epsilon$, where ϵ is very small. A small frictionless puck of thickness 2ϵ is inserted between the two cylinders, so that it can be considered a point mass that can move freely at a fixed distance from the vertical axis. then ρ is fixed at $\rho = R$, while ψ and z can vary at will. Write down and solve Newton's second law for the general motion of the puck, including the effects of gravity. Describe the puck's motion.

Problem 50

The differential equation (1.51) for the skateboard of Example 1.2 cannot be solved in terms of elementary functions, but is easily solved numerically.

1. If you have access to software, such as Mathematica, Maple, or Matlab, that can solve differential equations numerically, solve the differential equation for the case that the board is relased from $\psi_o = 20$ degrees, using the values $R = 5m$ and $g = 9.8m/s^2$. Make a plot of ψ against time for two or three periods.
2. On the same picture, plot the approximate solution (1.57) with the same $\psi_o = 20$ degrees. Comment on your two graphs.

Problem 51

Repeat all of Problem 1.50 but using the intial value $\psi_o = \pi/2$.